

Generic Upgrade Tool – Troubleshooting & Operations Guide

Fictional companion document for GraphRAG testing. All names, environments, incidents, and procedures are fictional.

1. Upgrade Job States

State	Meaning	Support Action
Pending	Upgrade request created but not started.	Verify approval and maintenance window.
Precheck Failed	One or more prerequisites failed.	Review the failed check and correct the prerequisite.
Ready	All prechecks passed.	Proceed during the approved window.
Running	Upgrade execution is in progress.	Monitor the job and avoid duplicate execution.
Validation Failed	Upgrade finished but post-upgrade checks failed.	Review service and application logs.
Completed	Upgrade and validation completed successfully.	Record final version and close the request.

2. Diagnostic Checklist

- Confirm the affected client and environment.
- Confirm current and target application versions.
- Check the upgrade job state.
- Review the exact error before retrying.
- Check Upgrade Manager and Execution Agent logs.
- Verify application service status.
- Verify database connectivity when required.
- Check disk space and package availability.
- Confirm the package is approved for the target version.
- Record all troubleshooting actions.

3. Scenario: Upgrade Package Not Found

Symptoms: The job reports that the requested package is unavailable. Checks: target version, repository connectivity, repository path, and package approval. Resolution: use the approved package; do not substitute an unapproved package. Escalation: provide job ID, target version, repository error, and timestamps.

4. Scenario: Execution Agent Offline

Symptoms: Upgrade Manager cannot communicate with the Execution Agent. Checks: agent service, host connectivity, configuration, and logs. Resolution: restore the agent service or correct approved configuration, then rerun the relevant precheck. Escalation: provide agent logs, host information, job ID, and last successful communication time.

5. Scenario: Validation Failed After Upgrade

Symptoms: Package execution completed but validation failed. Checks: application services, database connectivity, authentication, health checks, and logs. Resolution: correct the failed dependency and rerun

validation when permitted. Escalation: provide target version, failed check, logs, and timestamp.

6. Scenario: Database Migration Failure

Symptoms: Database migration stops during upgrade. Checks: connectivity, permissions, migration state, and database service. Resolution: do not repeatedly rerun; preserve logs and follow the database recovery procedure. Escalation: include migration error, database status, versions, and job ID.

7. Scenario: Application Unavailable After Upgrade

Symptoms: Execution completed but users cannot access the application. Checks: services, dependencies, ports, authentication, database connectivity, and logs. Resolution: restore the failed dependency and perform post-upgrade health checks. Escalation: capture user-facing error, service status, logs, target version, and failure time.

8. Support Evidence to Collect

- Client and environment.
- Upgrade job ID.
- Current and target versions.
- Start and failure timestamps.
- Exact error message.
- Upgrade Manager logs.
- Execution Agent logs.
- Application/database logs when relevant.
- Precheck or validation result.
- Troubleshooting steps already attempted.

9. Relationships Useful for Graph Search

- Upgrade Job → HAS_STATE → Job State
- Upgrade Job → RUNS_ON → Environment
- Upgrade Job → USES_PACKAGE → Upgrade Package
- Execution Agent → RUNS_ON → Environment
- Validation Check → VALIDATES → Application
- Database Migration → MODIFIES → Database
- Incident → AFFECTS → Environment
- Incident → RELATED_TO → Upgrade Job

10. Example GraphRAG Questions

- What should I collect before escalating an offline execution agent?
- What are the checks for a validation failure after an upgrade?
- What information is needed when a database migration fails?
- What state indicates that all prechecks have passed?

- What should be checked when an application is unavailable after an upgrade?
- What is the relationship between an upgrade job and an execution agent?
- What should a support engineer do when the requested package is unavailable?